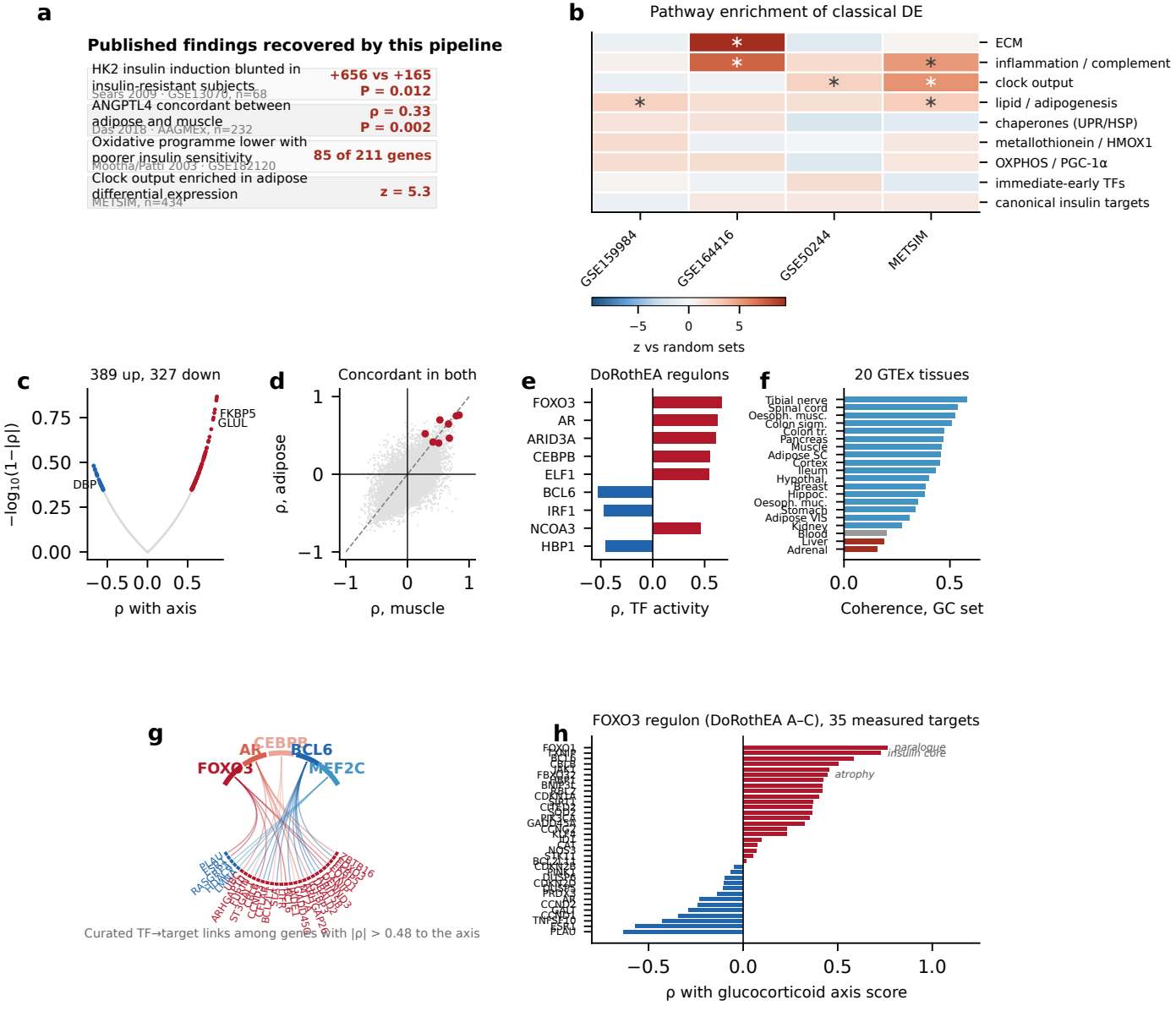

a
Published findings recovered by this pipeline
HK2 insulin induction blunted in insulin-resistant subjects
Sears 2009 · GSE13070, n=68
+656 vs +165
P = 0.012
ANGPTL4 concordant between adipose and muscle
Das 2018 · AAGMEX, n=232
ρ = 0.33
P = 0.002
Oxidative programme lower with poorer insulin sensitivity
Mootha/Patti 2003 · GSE182120
85 of 211 genes
Clock output enriched in adipose differential expression
METSIM, n=434
z = 5.3
b
Pathway enrichment of classical DE
ECM
inflammation / complement
clock output
lipid / adipogenesis
chaperones (UPR/HSP)
metallothionein / HMOX1
OXPHOS / PGC-1α
immediate-early TFs
canonical insulin targets
GSE159984
GSE164416
GSE50244
METSIM
−5
0
5
z vs random sets
c
389 up, 327 down
FKBP5
GLUL
DBP
−log10(1−|ρ|)
ρ with axis
d
Concordant in both
ρ, adipose
ρ, muscle
e
DoRothEA regulons
FOXO3
AR
ARID3A
CEBPB
ELF1
BCL6
IRF1
NCOA3
HBP1
ρ, TF activity
f
20 GTEx tissues
Tibial nerve
Spinal cord
Oesoph. musc.
Colon sigm.
Colon tr.
Pancreas
Muscle
Adipose SC
Cortex
Ileum
Hypothal.
Breast
Hippoc.
Oesoph. muc.
Stomach
Adipose VIS
Kidney
Blood
Liver
Adrenal
Coherence, GC set
g
AR
CEBPB
BCL6
FOXO3
MEF2C
Curated TF→target links among genes with |ρ| > 0.48 to the axis
h
FOXO3 regulon (DoRothEA A–C), 35 measured targets
paralogue
insulin core
atrophy
ρ with glucocorticoid axis score